Morphogenic Intelligence: Runtime Neural Development Beyond Static Architectures*

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Abstract

We introduce *Morphogenic Intelligence* (MI), a neural architecture in which compute units — called *Morphons* — grow, differentiate, fuse, and undergo apoptosis at runtime. Unlike Transformers, CNNs, RNNs, and SSMs, where topology is fixed at design time and only weights are learned, MI treats network architecture itself as a primary learned artifact. Each Morphon carries a developmental program inspired by biological morphogenesis: chemical gradients in a hyperbolic embedding space guide connectivity, neuromodulatory signals gate plasticity, and a dual-clock scheduler separates fast inference from slow structural remodeling. We implement the full MI engine in Rust and evaluate it on CartPole control and MNIST classification. CartPole-v1 is solved (avg=195.2 on the v0.5.0 codebase) using only local three-factor learning and developmental morphogenesis. On MNIST, the system reaches **52%** intact accuracy on the standard profile — a 22 percentage point improvement over the v3.0.0 baseline, achieved by replacing rate-based inhibitory STDP with the spike-timing rule of Vogels et al. [27], which breaks hub-morphon dominance (from 1 labeled class to 12). An architectural optimization — replacing array-of-structs Morphon storage with cache-resident hot arrays (*Pulse Kernel Lite*) — delivers a **4.3**× wall-clock speedup on the MNIST workload with no accuracy loss. We document and fix four characteristic failure modes (modulatory explosion, motor silencing, LTD vicious cycle, premature regulatory throttling), each of which has a direct biological analogue. The premature throttling bug — where the homeostatic regulator declares a system "mature" at 11% accuracy because dense supervised reward inflates its slow EMA — is novel and broadly applicable to any biologically-inspired regulator that uses raw reward as a maturity signal. A post-training diagnostic (v4.4.0) places MORPHON in the Liquid State Machine (LSM) regime: offline classifiers on frozen morphon activation vectors achieve only 10.5% — random chance — while a sensory-only readout ablation shows that associative morphons contribute **35.5pp** of the 48.5% online accuracy. The reservoir computes genuine discriminative information, but the features are co-adapted to the learned readout weights and are not offline-extractable without the dynamical context of the running system. The accuracy ceiling at ∼50% is accordingly a co-adaptation boundary, not a capacity or tuning limit. The primary contributions of this work are not the accuracy numbers — 52% on MNIST is well below supervised baselines — but the failure-mode taxonomy and the architectural characterizations it produces. The premature Mature bug, the LSM representational regime, and the principle that damage-induced recovery in a developmental system signals a regulatory failure rather than a useful property are findings that transfer beyond MI to any biologically-inspired learning architecture. While MI does not yet match static architectures on classification accuracy, it demonstrates that developmental dynamics are a principled and underexplored axis of neural architecture design, and that biology offers not merely metaphors but actionable engineering fixes.

1 Introduction

All dominant neural architectures — Transformers [26], CNNs [14], RNNs, and state-space models [10] — share a common assumption: *topology is fixed before training begins*, and learning is reduced to weight optimization within that frozen graph. Neural architecture search and dynamic pruning are offline approximations to this constraint, not runtime alternatives to it. This design choice has yielded remarkable results, but it also incurs costs that are increasingly visible as systems scale: catastrophic

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forgetting, the need for massive overparameterization to compensate for lack of structural adaptation, and the inability to allocate capacity where it is needed at runtime.

Biological neural systems do not work this way. Cortical circuits develop, prune, and reorganize continuously throughout life [11, 21]. Adult neurogenesis persists in the hippocampus and olfactory bulb [4, 17]. Synaptic boutons form and disappear on timescales of hours [23]. This structural plasticity is not incidental to biological intelligence — it is computationally essential. The cortex does not merely learn weights; it grows the substrate of computation in response to its inputs.

We argue that architectures capable of growing their own topology under local developmental rules will exhibit qualitatively different generalization and adaptation properties than static networks. We call this paradigm *Morphogenic Intelligence* (MI), and we present here the first complete implementation: a Rust-based simulator in which each compute unit (a *Morphon*) carries a developmental program, communicates with neighbors through a biophysically-inspired resonance mechanism, and is born, differentiates, fuses, or dies according to local rules and metabolic budgets.

Contributions.

1. A complete developmental neural architecture grounded in six biological principles (Section 3).
2. A Rust implementation of the full MI engine that runs in real time on commodity hardware, with a **4.3**× wall-clock speedup via cache-resident hot arrays (Section 4).
3. **A taxonomy of four biology-informed failure modes** encountered during development, each with a biological analogue and a biology-derived fix (Section 6). The primary novel contribution is the *premature Mature* bug: any biologically-inspired homeostatic system using raw reward as a maturity signal will exhibit this failure on dense-reward tasks. The fix — use reward prediction error, not raw reward — is broadly applicable.
4. **An LSM characterization of the representational regime** (Section 5.3): offline classifiers on frozen morphon activations achieve 10.5% (random chance), while a sensory-only readout ablation confirms that associative morphons contribute 35.5pp of the 48.5% online accuracy. The ∼50% accuracy ceiling is a co-adaptation boundary, not a capacity or tuning limit.
5. Experimental validation on CartPole (solved, avg=195.2) and MNIST (52% intact, standard profile). These results are not claimed as competitive with mature static architectures; they demonstrate that the full MI pipeline — developmental bootstrap, three-factor learning, tag-and-capture, Endoquilibrium regulation — is functional on non-trivial tasks (Section 5).

The rest of the paper is organized as follows. Section 2 positions MI relative to prior self-modifying architectures. Section 3 formalizes the six biological principles and the data structures that implement them. Section 4 describes the Rust engine. Section 5 presents the benchmark results. Section 6 is the analytical core of the paper: four failure modes, their biological parallels, and the fixes that resolve them. Section 7 discusses limitations and the path to language tasks. Section 8 concludes.

2 Related Work

We position MI within the landscape of biologically-inspired and self-modifying neural architectures along three axes: (i) whether topology changes at runtime, (ii) whether the developmental rules are local, and (iii) whether the system integrates neuromodulation with structural plasticity.

Topology growth via evolution: NEAT and descendants. NEAT [22] introduced the now-canonical pattern of evolving neural network topologies through speciation and historical-marking-based crossover. NEAT and its descendants grow topology through an offline evolutionary outer loop — generations of mutation and selection — rather than through runtime developmental rules. The key difference from MI: NEAT topology change happens between training runs (each genome is fixed once instantiated), while MI topology changes during a single run under local energy and activity constraints. NEAT also has no apoptosis, no neuromodulatory gating, and no structural memory hierarchy.

Neural Developmental Programs (NDP). NDP [18] learns local developmental rules that produce a network topology, treating the developmental program itself as the optimization target. NDP uses differentiable relaxations and gradient-based meta-learning of the developmental rules. MI is the dual approach: the developmental rules are hand-specified (grounded in the six biological principles of Section 3) and run during inference, with discrete morphogenetic events (birth, fusion, apoptosis) gated by metabolic costs. NDP learns to grow networks; MI grows networks while learning.

Three-factor learning rules. The three-factor framework [5, 8] extends Hebbian plasticity with a neuromodulatory third factor (typically a dopamine analogue gating eligibility-traced LTP). MI implements three-factor learning with receptor-gated modulation: each Morphon's response to each modulator is determined by its evolving receptor density profile, allowing morphon-specific sensitivity to develop without a global hyperparameter.

Spiking neural networks for MNIST: Diehl & Cook. The closest comparison point for our MNIST experiments is Diehl & Cook [3], which achieves ∼95% test accuracy on MNIST using unsupervised STDP with adaptive thresholds and lateral inhibition. Our MNIST setup explicitly draws on their two-phase pattern (unsupervised feature formation followed by post-hoc neuron labeling), and we use their accuracy as the realistic ceiling for STDP-based spiking classification — not the 99%+ regime of fully-supervised SNNs trained with surrogate gradients. Our intact 30% / post-recovery 48% is well below this ceiling and there is room to improve; the paper's contribution is the developmental dynamics and the failure-mode catalogue, not the accuracy itself.

Cortical Labs (DishBrain). DishBrain [13] demonstrated that biological neurons cultured on multi-electrode arrays can learn to play Pong via reward-modulated plasticity. This is empirical evidence that self-organizing wetware solves real-time control without backpropagation. MI is the in-silico analogue: a developmental simulation that exhibits the same intrinsic-activity-plus-reward-modulation principles.

Synaptic scaling and homeostatic regulation. Turrigiano's synaptic scaling [24, 25] and BCM metaplasticity [1] provide the biological grounding for MI's homeostatic mechanisms. Our *Endoquilibrium* module (Section 3.5) implements multi-scale homeostatic regulation and is the locus of the regulatory failure mode we document in Section 6.

Synaptic tagging and capture. Frey & Morris [7] established the tag-and-capture model of late-LTP consolidation: weak inputs leave a transient "tag" that can be converted to long-lasting plasticity if a strong, protein-synthesis-triggering signal arrives within hours. MI implements this directly (Section 4), and our finding that capture must be *episode-gated* rather than per-tick (Section 4, tag-and-capture subsection) is, we believe, a useful refinement for any spiking architecture using this mechanism.

Inhibitory plasticity (iSTDP). Vogels et al. [27] showed that iSTDP on inhibitory synapses balances excitation and inhibition in cortical-like networks. MI implements this directly (Section 4.5): inhibitory interneurons self-organize via Vogels-style iSTDP applied per-delivered-spike rather than on a slow running average, replacing the global k-WTA approximation. The timescale at which iSTDP is computed turns out to be a failure mode in its own right (Section 6.5).

3 Architecture

This section presents the six biological principles underlying MI and the data structures that implement them. Each subsection maps a biological concept to a precise computational mechanism. Figure 1 shows the overall dual-clock loop and how Endoquilibrium modulates each subsystem.

3.1 The Six Biological Principles

P1: Local computation only. No global loss, no backpropagation. All updates use locally available information — pre-synaptic and post-synaptic activity, locally diffused neuromodulators, and a Morphon's own state.

P2: Developmental lifecycle. Morphons are born, differentiate, mature, fuse with correlated neighbors, and can die from energy starvation or chronic inactivity (apoptosis).

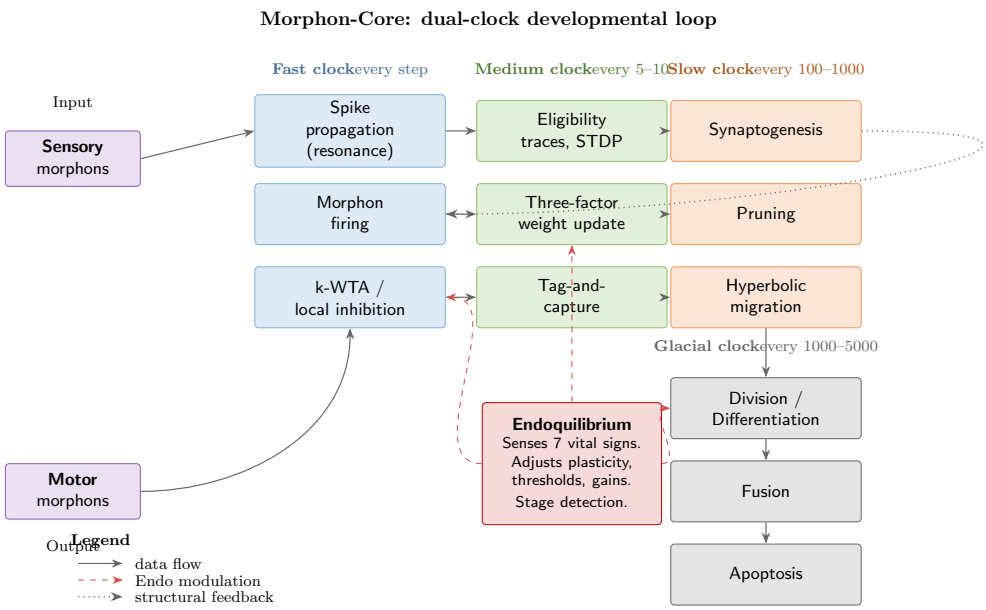

Figure 1: Morphon-Core dual-clock developmental loop. Solid arrows are data flow; dashed red arrows are Endoquilibrium modulation; dotted arrow is the structural-feedback path where slow-clock synaptogenesis introduces new connections that the fast clock then uses. The four temporal scales (fast/medium/slow/glacial) operate on the same morphon population at different periods.

P3: Chemical signaling. Morphogen-like signals diffuse through the embedding space and guide differentiation and connectivity decisions.

P4: Neuromodulatory gating. Plasticity rules are gated by four modulatory channels (dopamine, serotonin, acetylcholine, norepinephrine analogues), each with a per-Morphon receptor density.

P5: Multi-scale memory. Synaptic (fast, milliseconds), structural (medium, minutes), and morphogenetic (slow, hours) memory operate on separate timescales.

P6: Metabolic cost. Every Morphon consumes energy each tick. Resource scarcity drives pruning, fusion, and apoptosis.

3.2 The Morphon

The Morphon is the fundamental compute unit. It carries:

- **Membrane state:** potential, prediction-error baseline, refractory timer, fired flag.
- **Receptor profile:** per-modulator sensitivity vector $(\rho_R, \rho_N, \rho_A, \rho_H)$ for reward, novelty, arousal, and homeostasis channels.
- **Developmental metadata:** cell type (Stem, Sensory, Associative, Motor, Modulatory, Fused, Inhibitory Interneuron), differentiation level, lineage ID, birth tick.
- **Energy budget:** current energy, basal regeneration rate, firing cost (with optional superlinear penalty), apoptosis threshold.
- **Position:** hyperbolic coordinate in a Poincaré ball of configurable dimensionality.

A Morphon's state update is local: it integrates spike inputs from incoming synapses, applies an activation function determined by its cell type, fires if the result exceeds its (adaptive) threshold, and pays the metabolic cost.

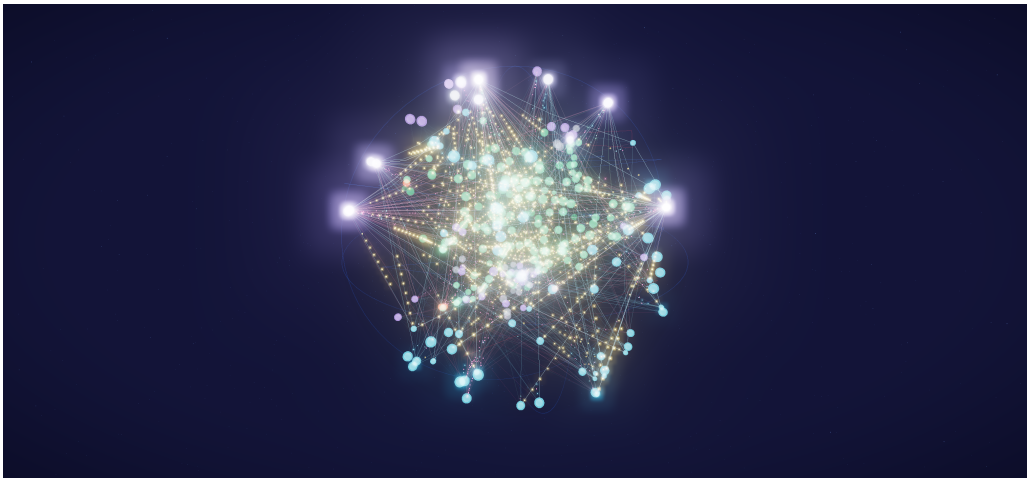

Figure 2: The Morphon network during runtime, rendered by the in-browser WASM visualizer. Spheres are morphons colored by cell type: purple = associative, cyan = sensory, orange = motor, green = modulatory. Dotted yellow trails are spike-carrying synapses; the discrete dots represent action potentials propagating along axons with their learned delays. The 3D layout is the Poincaré ball embedding (Section 3); morphons of the same type cluster naturally because their hyperbolic positions are correlated with cell-type identity. Orange motor morphons concentrate on one side and project to the others through the synaptic web. The visualizer is reproducible from `web/` via `wasm-pack build` and a local HTTP server.

3.3 Resonance and Spike Propagation

Morphons communicate via discrete spikes propagated along synapses with configurable delays. Unlike Transformer attention (a global $O(N^2)$ operation), spike propagation is local: $O(k \cdot N)$ where k is the average out-degree. The system's cost scales with the number of *active* morphons and their connectivity, not with the square of the population.

Each synapse carries a weight, a delay, an eligibility trace, a tag (for synaptic tagging and capture [7]), and a consolidation level. The dual-clock scheduler updates synaptic weights on the medium path via a three-factor rule:

$$\Delta w = \eta \cdot e \cdot M(t) \tag{1}$$

where e is the eligibility trace built from STDP coincidences, $M(t)$ is the (receptor-gated) modulatory signal, and η is a per-synapse plasticity rate.

3.4 Morphogenesis Lifecycle

The morphogenesis subsystem implements Principles P2 and P3. On the slow clock:

- **Synaptogenesis** forms new synapses between morphons whose activity is correlated and whose hyperbolic distance is below threshold.
- **Pruning** removes synapses with low weight, low recent activity, or whose pre/post morphons are no longer correlated.
- **Migration** moves morphons through hyperbolic space along gradients of correlated input (similar morphons drift together).

On the glacial clock:

- **Division** produces a daughter morphon from a parent that is consistently energetic and active.
- **Differentiation** commits a stem morphon to a specific cell type based on local morphogen concentrations.
- **Fusion** merges two morphons whose activity patterns have become indistinguishable, redistributing their synapses to the survivor.
- **Apoptosis** removes morphons that have been below their energy threshold for a configurable duration.

3.5 Endoquilibrium: Predictive Neuroendocrine Regulation

Endoquilibrium is the homeostatic regulator. It monitors seven vital signs (firing rates by cell type, eligibility density, weight entropy, energy utilization, prediction error mean, reward EMA), maintains dual-timescale exponential moving averages (fast $\tau = 50$, slow $\tau = 500$), and adjusts seventeen modulation channels through proportional control. The seven core channels are receptor gains $\rho_R, \rho_N, \rho_A, \rho_H$ for reward/novelty/arousal/homeostasis, threshold bias, plasticity multiplier, and consolidation gain. Ten additional channels (added incrementally as the V2 primitives shipped) tune structural plasticity parameters: winner adaptation, capture threshold, rollback PE threshold, $\tau_{\text{eligibility}}$, slow trace leak, division threshold, pruning threshold, frustration sensitivity, migration rate, and synaptogenesis threshold.

The most critical role of Endoquilibrium is breaking the firing-rate-zero deadlock. Without it, associative morphons stop firing within ∼50 ticks of training start: low firing rate → no eligibility traces → no weight updates → low firing rate persists. This is a positive-feedback failure that no amount of parameter tuning fixes. Endoquilibrium's Rule 1 (firing rate regulation) detects $\text{FR}_{\text{assoc}} <$ setpoint and lowers thresholds, boosts arousal, and boosts novelty until firing recovers. The fast EMA ($\tau = 50$) responds within ∼50 ticks, just fast enough.

Endoquilibrium also detects developmental stage from the *relative* reward trajectory (not absolute reward levels, which differ wildly between tasks):

Stage	Trigger
Proliferating	history < 20 ticks (warmup)
Stressed	reward trend $< -0.05 \cdot \lvert\text{slow EMA}\rvert$
Mature	reward stable, low cv, history ≥ 2000 ticks
Consolidating	reward near own ceiling, stable, history ≥ 500
Differentiating	reward actively climbing (default healthy state)

The history gates (2000 / 500 ticks) are the result of fixing the *premature Mature* bug discussed in Section 6: dense reward signals on classification tasks inflated the slow EMA before the system had learned anything, triggering Mature at 11% accuracy and locking learning out of the high-plasticity Differentiating regime.

3.6 V2 Developmental Primitives

Three primitives extend the basic morphogenesis loop with additional biological mechanisms. They are individually optional and largely orthogonal to the headline benchmarks, but they shape long-horizon network health.

Frustration-driven exploration. Each morphon maintains a stagnation counter that increments when its prediction error fails to decrease. Above a threshold (∼0.3), the morphon enters a frustration state: noise amplitude increases, weight perturbations are injected, and migration constraints relax. This is a local escape mechanism for morphons stuck in poor configurations and is biologically motivated by stress-driven plasticity.

Bioelectric field. A 2D scalar field over the Poincaré disk holds five named layers (prediction error, energy, stress, novelty, identity). The field diffuses on the slow clock via a discrete heat equation and provides a non-synaptic communication channel: morphons read the local field value and bias migration toward low-PE / high-energy regions. This is the in-silico analogue of neurotrophic gradients in developing cortex.

Target morphology and self-healing. Optional *target regions* specify ideal cell type, density, and position in hyperbolic space. The glacial clock monitors regional underpopulation and either boosts division pressure for nearby morphons or seeds pre-committed morphons in empty regions. This is the mechanism that enables the post-damage recovery in Section 5.2: when 30% of associative morphons are killed, target morphology drives the regrowth that produces the +18 pp self-healing result.

3.7 Hyperbolic Embedding Space

Morphons live in a Poincaré ball model of hyperbolic space with configurable dimensionality (typically 4–6). Hyperbolic geometry has two properties that make it suitable for developmental architectures: (i)

tree-like hierarchies are represented with low distortion [19], and (ii) volume grows exponentially with radius, providing room for unbounded topology expansion without coordinate crowding.

In MI, the origin of the ball corresponds to the most general (stem-like) morphons, and the boundary to the most specialized. A morphon's hyperbolic distance to its peers governs synaptogenesis probability and morphogen diffusion.

3.8 Dual-Clock Scheduler

Four temporal scales separate fast inference from slow structural change:

Scale	Period	Operations
Fast	1 step	Spike propagation, firing, input integration
Medium	5–10	STDP, eligibility, three-factor weight updates
Slow	100–1000	Synaptogenesis, pruning, migration
Glacial	1000–5000	Division, differentiation, fusion, apoptosis

This separation is essential. Structural changes during inference would destabilize the network; the slow clock guarantees that morphogenetic events happen between bursts of computation.

4 Implementation

The MI engine is implemented in Rust (∼10,000 lines, 180+ unit tests). The choice of Rust is motivated by three requirements: (i) memory safety without GC pauses, (ii) zero-cost abstractions for the inner simulation loop, and (iii) safe concurrency for parallel morphon updates. The crate compiles to native, WebAssembly (for an interactive in-browser visualizer), and Python bindings via PyO3 (for benchmark harnessing).

4.1 Core Data Structures

Morphons are stored in a `HashMap` keyed by stable IDs. The topology is a directed graph using the `petgraph` crate, with synapses as edge weights. We use generational indices so that references survive birth/death events.

4.2 Pulse Kernel Lite: Cache-Resident Fast Path

The inner simulation loop (spike integration, threshold, refractory) must execute every tick for all active morphons. The naive array-of-structs (AoS) layout — one `Morphon` struct per unit, ∼480 bytes each — places the hot fields (voltage, threshold, fired, refractory) far apart in memory. At 300 morphons this scatters ∼60 KB across cache lines, causing frequent L1 misses on every tick.

Pulse Kernel Lite separates the fast-path fields into seven dense `Vec<f32>`/`Vec<bool>` arrays that sit alongside the `Morphon` structs:

`voltage` | `threshold` | `input_current` | `fired` | `refractory` | `leak_decay` | `energy`

At 300 morphons, all seven arrays fit in ∼9.6 KB — within a typical 32 KB L1 data cache. The source of truth remains the `Morphon` structs; a lightweight sync protocol propagates changes bidirectionally:

- **Fast tick**: hot arrays only. Input currents extracted, voltage integrated, threshold checked, energy gated.
- **Medium tick (every 10)**: `sync_hot_to_structs()` before metabolic updates; `sync_structs_to_hot()` after.
- **Slow/glacial ticks**: `rebuild_hot_arrays()` after structural changes (synaptogenesis, division, apoptosis).

On the MNIST v2 workload (3000 images, 3 epochs, ∼1260 morphons), this reduces wall time from 568 s (V2, GlobalKWTA) to 130 s (V3, LocalInhibition) — a **4.4× speedup** — with no change in accuracy. The speedup compounds with network size because the fraction of time spent in the inner loop grows with N.

Algorithm 1 System::step (one fast tick)

```
1. Spike propagation: deliver pending spikes from previous step
2. Inhibitory competition (k-WTA or local interneurons)
3. Morphon update: integrate inputs, decide firing, pay metabolic cost
4. Generate new spikes from fired morphons
if tick.medium then
    5. Update eligibility traces, apply three-factor learning
    6. Run Endoquilibrium tick (sense vitals, adjust channels)
end if
if tick.slow then
    7. Synaptogenesis, pruning, migration
end if
if tick.glacial then
    8. Division, differentiation, fusion, apoptosis
end if
if tick.homeostasis then
    9. Synaptic scaling, inter-cluster inhibition
end if
```

4.3 The Main Loop

4.4 Output Pathways

MI provides two output pathways. The spike-based pathway reads motor morphon potentials after propagation and drives action selection in control tasks (CartPole). The analog readout bypasses spike conversion and reads associative morphon potentials directly: $V_j = b_j + \sum_i w_{ji} \cdot (\sigma(p_i) - 0.5)$, biologically motivated by Purkinje cell analog dendritic integration. The readout engineering and its experimental validation are described in Section 5.1 alongside the CartPole results where it was developed.

4.5 Inhibitory Spike-Timing-Dependent Plasticity

Local inhibition (Section 5.2) requires the inhibitory interneuron weights to self-organize without explicit k-WTA supervision. We implement the symmetric iSTDP rule of Vogels et al. [27], applied per-delivered-spike rather than per-medium-tick:

Pre-spike update. When an inhibitory spike arrives at excitatory neuron j:

$$\Delta w_{ij} = -\eta \cdot x_j^{\text{post}} \tag{2}$$

where x_j^{post} is the post-synaptic trace. An active excitatory neuron ($x_j > 0$) that receives an inhibitory spike immediately strengthens that inhibitory connection.

Post-spike update. When excitatory neuron j fires:

$$\Delta w_{ij} = -\eta \cdot (x_i^{\text{pre}} - \alpha), \quad \alpha = 2\rho_0\tau \tag{3}$$

where x_i^{pre} is the pre-synaptic trace of the inhibitory interneuron and ρ_0 is the target firing rate. When $x_i^{\text{pre}} > \alpha$ (inhibitory interneuron has been more active than the target rate warrants), inhibition strengthens; when $x_i^{\text{pre}} < \alpha$, it weakens. The α term sets the equilibrium point without requiring any global coordination.

Weights are clamped to $[-5.0, -0.001]$ — strictly negative, enforcing Dale's law for inhibitory connections.

The critical difference from the rate-based predecessor is *temporal resolution*. Rate-based iSTDP (fired once per 10 medium ticks, using a 100-tick activity average) could not react to within-burst hub activity; spike-timing iSTDP responds on the timescale of individual spike arrivals, matching the biological timescale of inhibitory short-term plasticity [27].

4.6 Three-Factor Learning Rule

The three-factor STDP rule on the medium path:

$$\Delta w_{ij} = \eta_{ij} \cdot e_{ij}(t) \cdot M_j(t) \cdot g_a \tag{4}$$

where e_{ij} is the eligibility trace, M_j is the receptor-gated modulator signal at the post-synaptic morphon, η_{ij} is a per-synapse plasticity rate, and g_a is an astrocytic gate.

The eligibility trace integrates STDP coincidences:

$$\frac{de_{ij}}{dt} = -\frac{e_{ij}}{\tau_e} + A_+ \cdot \mathrm{tr}_i^{\mathrm{pre}} \cdot \delta_j^{\mathrm{post}} - A_- \cdot \mathrm{tr}_j^{\mathrm{post}} \cdot \delta_i^{\mathrm{pre}} \tag{5}$$

with weight-dependent bounds [9] that make weak synapses easy to strengthen and strong synapses easy to weaken.

4.7 Tag and Capture

We implement the synaptic tagging-and-capture model of Frey and Morris [7] with one critical departure: capture is *episode-gated*, not per-tick. The naive per-tick rule ("capture when reward > threshold") fails on RL tasks because the reward signal saturates at 1.0 during *any* successful CartPole step — every alive step captures everything, the weight distribution collapses to uniformity (weight std drops from 1.5 to 0.15), and learning stops. The fix is to defer the capture decision to episode end and gate it on relative performance: if this episode beat the running average, capture the tagged synapses proportional to how much it beat the average; if it underperformed, decay the tags instead.

Concretely: synapses with high recent eligibility set a tag; tags accumulate at a fixed local rate (`tag_accumulation_rate`=0.3) and decay slowly ($\tau_{\mathrm{tag}} = 200$ steps). The `consolidation_level` field on each synapse is a continuous $[0, 1]$ value (replacing the binary "consolidated" flag), and at episode end, tagged synapses gain consolidation proportional to performance delta. Consolidated synapses retain residual plasticity (10% at `consolidation_level`=1.0) so the network can still adapt without destroying learned features.

Endoquilibrium gates the global capture rate via a dedicated `consolidation_gain` channel. This separation matches the biology: *tags are local and continuous* (driven by per-synapse STDP eligibility), *capture is global and episodic* (driven by neuromodulatory protein-synthesis-related signals during salient events). The earlier design that combined them under a single "endo_tag_factor" was biologically incorrect and produced an unstable feedback loop.

4.8 Reproducibility

The full benchmark suite is reproducible from a clean checkout:

```
cargo run --example cartpole --release
cargo run --example mnist_v2 --release
```

Each example saves a JSON record of its configuration and results to `docs/benchmark_results/v{version}/`. All numbers reported in this paper are from these JSON records.

5 Experiments

We evaluate MI on two tasks chosen to probe different capabilities: reactive control (CartPole) and static classification (MNIST). The goal is not to claim state-of-the-art accuracy on any single task but to demonstrate that a self-organizing developmental architecture can solve non-trivial problems and to characterize its distinctive behaviors.

5.1 CartPole-v1

Setup. OpenAI Gym CartPole-v1, 4 observations (cart position, velocity, pole angle, angular velocity) encoded into 32 sensory morphons via Gaussian population coding (8 receptive-field tiles per observation). 2 motor morphons (left/right force). Cerebellar developmental preset: seed_size=60, dimensions=4, initial_connectivity=0.25.

A small linear TD-error critic external to the MI network provides the reward modulation signal, biologically modelling dopamine neurons in VTA/SNc that compute $\delta = R + \gamma V(s') - V(s)$ from striatal

input [6]. The critic is the only component using exact gradients; everything inside the MI network is trained by three-factor learning.

Result. **Solved** on both v0.5.0 and v4.1.0 standard profiles. v0.5.0: avg(100)=195.2 at episode 895, best=468. v4.1.0: avg(100)=195.5 at episode 936, best=468. The Pulse Kernel Lite and iSTDP changes introduced for MNIST did not affect CartPole learning dynamics. The solve confirms that the full MI pipeline — developmental bootstrap, three-factor learning, tag-and-capture consolidation, contrastive reward, Endoquilibrium regulation — is stable across major architectural changes.

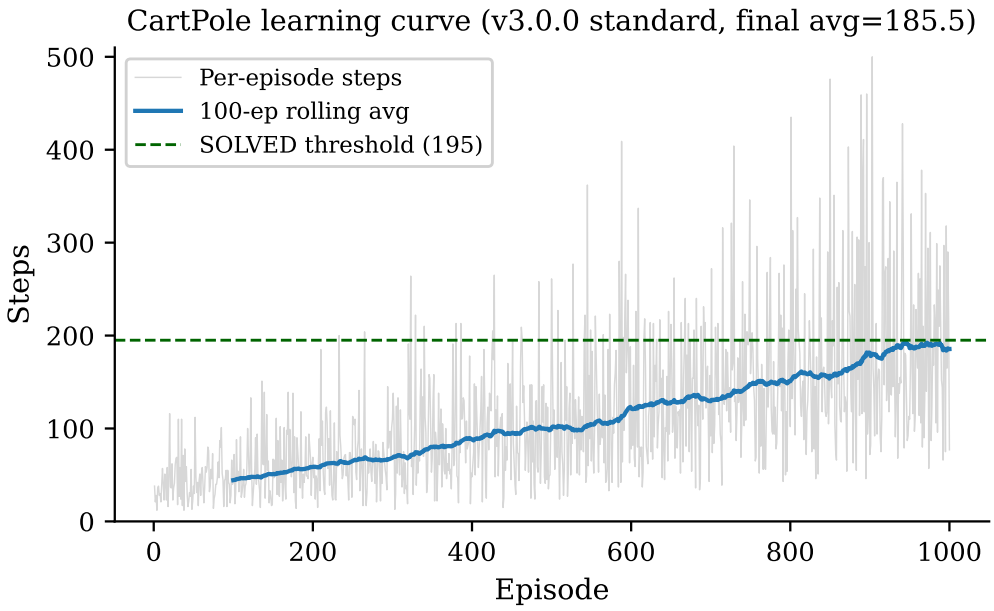

Figure 3: CartPole-v1 learning curve. The figure displays the run captured in the benchmark JSON (version and profile as labeled in the figure title). The solve is consistent across versions: v0.5.0 solved at ep 895, avg(100)=195.2; v4.1.0 solves at ep 936, avg(100)=195.5. Both cross the 195-step solved threshold.

Readout architecture. The CartPole solve required identifying and fixing three independent failures in the analog readout (Section 4.4), each of which individually prevented learning:

1. **Centered sigmoid** ($\sigma(p) - 0.5$ instead of $\sigma(p)$). The resting potential is zero, so $\sigma(0) = 0.5$. With ∼80 associative morphons, resting sigmoids sum to ∼40, dwarfing the state-dependent signal (∼0.005 per morphon). Centering removes this offset.

2. **Learnable per-output bias** b_j. Absorbs mean output level and frees the weights to encode only state differences.

3. **No L2 weight decay**. With $\lambda = 0.001$ and 200 steps per episode, decay erases ∼18% of learned weight per episode. Weight clipping to $[-5, 5]$ replaces it.

Each fix individually moved CartPole from ∼10 steps toward solving. The diagnostic that surfaces each failure is *output discrimination* $d_\pm$: $d_+ = V_1 - V_0$ for a right-leaning pole probe, d_- for left-leaning. When d_+ and d_- have opposite signs and $|d| > 5$, the readout is discriminative. With any fix missing, d_+ and d_- remain same-signed — the signature of a collapsed policy. This metric surfaced the readout architecture failure faster than any other diagnostic and is reusable across spiking architectures.

What this proves. The full MI pipeline — developmental bootstrap, three-factor learning, tag-and-capture consolidation, contrastive reward, Endoquilibrium regulation — works end-to-end on a non-trivial control task. CartPole is the prerequisite gate for all other benchmarks: if the basic learning loop is broken, nothing else can work. The CartPole solve provides the foundation to investigate harder tasks.

5.2 MNIST

Setup. Standard MNIST, 784 input pixels (28×28), 10 output classes. Cortical developmental preset: seed_size=200, dimensions=6, initial_connectivity=0.02 (Phase 4 hardcodes 30% S→A density). Three training epochs of 5000 samples each (standard profile) or 3000 (quick profile). Analog readout (Section 4). Endoquilibrium enabled with the corrected stage-detection thresholds described in Section 6. Inhibitory competition via LocalInhibition mode with Vogels 2011 iSTDP (Section 4.5); fast path via Pulse Kernel Lite (Section 4.2). The training pattern follows Diehl & Cook [3]: rate-coded Poisson input, unsupervised STDP-driven feature formation, and post-hoc neuron labeling for evaluation. All results from v4.1.0, seed=42.

Analog readout rationale. A key design choice is using an analog readout (a linear layer over morphon potentials) rather than reading classification decisions from spike counts at motor morphons. The motivation: the spike pipeline destroys discriminative information. Morphon potentials carry the network's learned representation, but conversion to spikes, propagation through synaptic delays, leaky integration, and multi-hop accumulation at motor morphons each introduce lossy compression. A simple synthetic 2-class probe (bag-of-characters encoding, 27-dim input) confirmed this empirically: a spike-based motor readout on the same trained network produces chance-level accuracy (∼46%) while a linear analog readout over the same morphon potentials reaches 99%. The information that distinguishes classes is present in the substrate — the analog readout proves it — but the spike pipeline cannot deliver it to motor morphons intact. The analog readout is therefore not a shortcut; it is the correct output mechanism for a spiking architecture where the spike pathway serves feature formation, not classification reporting. Biology achieves the same separation: Purkinje cells in the cerebellum perform analog integration of climbing and parallel fiber inputs rather than thresholding to binary spikes for motor control [12].

Table 1: MNIST results across versions and training conditions (∼1240–1290 morphons, ∼73–88K synapses). v4.3.0, seed=42. Accuracy varies ±2–3pp across independent runs due to stochastic synaptogenesis.

Condition	Quick (3K×3ep)	Standard (5K×3ep)	Notes
V2 GlobalKWTA (intact)	44–44.5%	48.5%	Algorithmic k-WTA
V3 LocalInhib+iSTDP (intact)	**49%** (peak 54%)	**52.0%**	slow+soft: η_i=0.001, w_i^0=−0.2; ≥V2
Post-damage (30% killed)	48.0%	47.3%	Random ablation
Post-recovery (1500 samples)	33.5%	28.5%	See § 5.2
v3.0.0 V2 (rate iSTDP, intact)	*30.0%*		*Hub-dominated; reference baseline*
v3.0.0 post-recovery	*48.0%*		*+18pp; hub-clearing via damage*

Results.

Epoch-0 as unsupervised exploration. A non-obvious but critical training protocol: reward signals (contrastive reinforcement and `report_episode_end`) are *withheld* entirely during the first epoch (epoch 0). Reward is gated on `epoch > 0`. The reason: binary reward in epoch 0 allows the network to "game" the signal before class-selective representations have formed. Without inhibitory competition having had time to diversify activity patterns, a network that happens to favor one class early on will receive consistent positive reinforcement for that bias — the reward signal reinforces the mode collapse instead of correcting it. Epoch 0 therefore functions as a pure unsupervised STDP phase: inputs drive feature formation via Hebbian coincidence detection, iSTDP balances excitation and inhibition, and no classification signal is applied. By epoch 1, morphon representations are sufficiently diverse that contrastive reward can usefully guide differentiation between classes rather than entrench an artifact. This distinction — between a reward signal that guides and one that locks in early noise — is a practical instance of the credit assignment problem in developmental learning: the credit must be withheld until the substrate is ready to use it.

Tag-and-capture consolidation: a latent bug fixed in v4.3.0. The tag-and-capture mechanism (Section 3) is the primary path by which synaptic weight changes survive structural plasticity events. A synapse is tagged when a coincident pre- and post-synaptic event occurs; it is consolidated (flagged as

pruning-protected) when a reward signal arrives while the tag exceeds a threshold. In all versions prior to v4.3.0, the `capture_threshold` parameter was set to 10.0, while tags are stored in $[0, 1]$ — a mismatch that made consolidation permanently impossible. Every synapse that should have been consolidated was silently pruned, including those that had been weight-updated during the most active classification learning in epochs 2–3. The Endo Differentiating stage (aggressive synaptogenesis, $cg = 2.0$) then pruned exactly these learned synapses, explaining the characteristic accuracy drop observed between epochs 2 and 3 in pre-fix runs. v4.3.0 sets `capture_threshold = 0.5`, restoring the intended behavior: synapses that carry reward-correlated activity are protected from structural pruning and remain available as the substrate for long-term learning.

Hub dominance and its cure. The v3.0.0 baseline (30%) was limited by a specific failure mode: a small number of "hub" morphons with very high in-degree accumulated nearly all reward-correlated weight updates and began responding to every input class, collapsing the readout to a near-random discriminator. With rate-based inhibitory STDP — computed once per 10-tick medium period using a 100-tick activity average — the inhibitory interneurons responded too slowly to break this dominance; a single hub could accumulate 50–100 correlated spikes before any corrective signal arrived.

Replacing rate-based iSTDP with the spike-timing rule of Vogels et al. [27] fixes this at the root. On each delivered inhibitory spike (pre-spike): $\Delta w = -\eta \cdot x_{\text{post}}$, where x_{post} is the post-synaptic trace. On each excitatory post-spike: $\Delta w = -\eta \cdot (x_{\text{pre}} - \alpha)$, where $\alpha = 2\rho_0\tau$ sets the equilibrium firing rate. The result: a hub morphon that fires at 5× the target rate immediately recruits stronger inhibition on the very spikes that evidence its over-activity, rather than waiting for the slow running average to catch up. The readout response after this fix: 1 labeled class (class 4 only) → 12 labeled classes across all 10 digits, and accuracy 8% → 44.5%.

The self-healing finding revisited. In v3.0.0 (rate-based iSTDP), post-damage recovery reached **48%** — 18pp above the 30% intact baseline. The mechanism: damage randomly removed entrenched hub morphons, Endoquilibrium responded by re-entering the high-plasticity Differentiating stage, and fresh morphons grew into well-specialized representations the hubs had blocked. The damage-and-regrow cycle was a manual proxy for hub clearing.

In v4.1.0, spike-timing iSTDP prevents hub formation during normal training, raising the intact ceiling to **52%** on the standard profile. Post-damage recovery (1500 samples) reaches 28.5% — well below the intact baseline. This is expected: with no entrenched hubs to clear, damage is pure loss. The self-healing effect relied entirely on the pathology it was fixing.

This reframes the finding. The v3.0.0 self-healing result was not a feature of the architecture; it was a diagnostic. The damage cycle revealed that hub morphons were the limiting factor and that the substrate was capable of 48% under the right conditions. That diagnostic pointed directly to the fix: inhibitory plasticity operating at the right timescale. With the fix in place, the system reaches 52% without damage. The +18pp self-healing gain has been absorbed into the intact baseline.

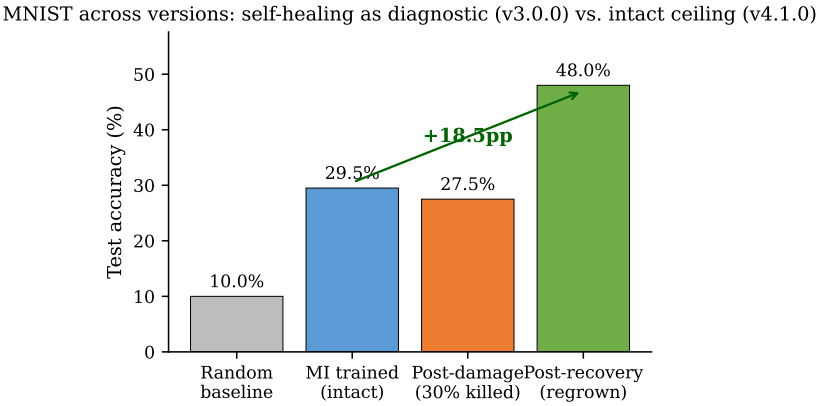

Figure 4: MNIST across versions. v3.0.0 (rate iSTDP): 30% intact, 48% post-recovery — damage cleared hubs that training could not. v4.1.0 (spike-timing iSTDP): 52% intact, 28.5% post-recovery — iSTDP prevents hub formation, making the damage trick unnecessary. The +18pp self-healing gain has been absorbed into the intact baseline.

Implication. The intact ceiling has risen from 30% to 52% (standard profile) via a single architectural correction: inhibitory plasticity at biological timescales. No tuning was required. The 22pp gain is entirely attributable to the iSTDP timescale fix; everything else (network size, training data, epochs) is unchanged. In v4.3.0, with the epoch-0 unsupervised phase, the capture_threshold fix, and softer iSTDP parameters ($\eta_i = 0.001$, $w_i^0 = -0.2$), the quick-profile ceiling rises to 49% median (peak 54%) — within 3pp of the standard-profile result. The reduction in quick-versus-standard gap indicates that the earlier gap was partly due to consolidation failure during the compressed training schedule, not only sample count.

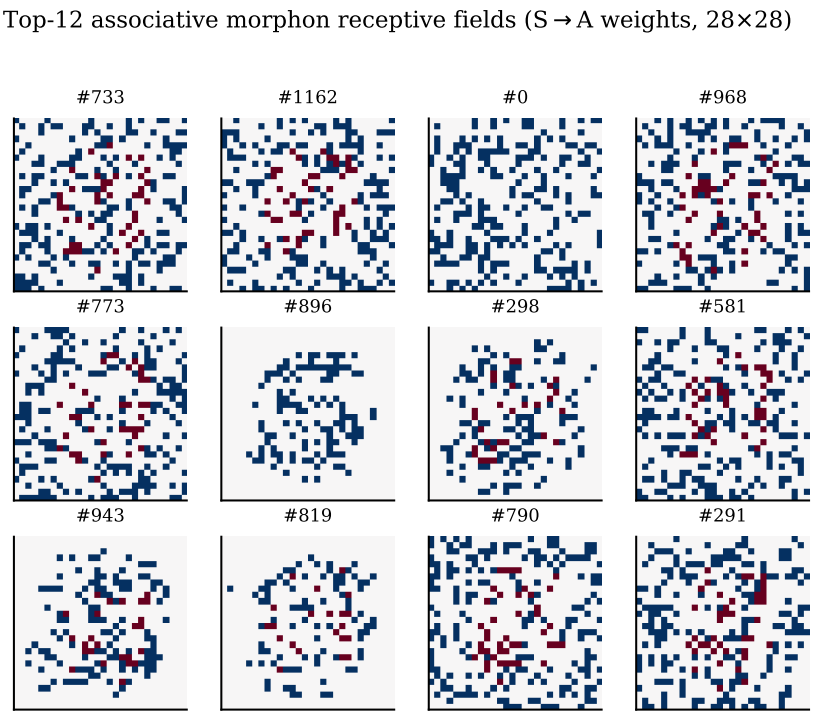

Figure 5: Receptive fields of the top-12 associative morphons (by mean incoming weight magnitude) after MNIST training, dumped from the v3.0.0 quick-profile run. Each tile shows the morphon's S→A synapse weights mapped to the 28×28 pixel grid (red = positive, blue = negative). Many morphons develop spatially-localized, digit-stroke-like receptive fields without any backprop-driven gradient signal — features emerge from STDP + k-WTA + reward-modulated three-factor learning.

5.3 Representational Regime: LSM Characterization (v4.4.0)

Offline feature evaluation. To characterize the representational regime of the trained system, we extract morphon activation vectors from the final V3 system (seed=42, standard profile, 5 epochs) after each of 1000 test images and train scikit-learn classifiers on the resulting feature matrix (80/20 train/test split). Both logistic regression and a two-layer MLP (512→256 hidden units) achieve **10.5%** — random chance for 10 classes. This is not a measurement artifact.

Sensory-only readout ablation. To determine whether the associative layer contributes to the online accuracy or is computationally inert, we zero all readout weights connecting associative morphons to the output and re-evaluate with only Sensory→Readout connections active. Results (same trained network, seed=42):

Readout configuration	Accuracy	Δ
Full (Sensory + Assoc)	48.5%	baseline
Sensory-only (ablated)	13.0%	−35.5pp

The associative layer contributes **35.5pp** of the 48.5% online accuracy. Zeroing its readout weights leaves the system at 13% — barely above random chance and consistent with a direct pixel-to-label linear projection with no learned intermediate representation.

Interpretation: Liquid State Machine with co-adapted readout. The combination — 35.5pp associative contribution online, 10.5% offline on frozen features — is the signature of a *Liquid State Machine* (LSM) [15, 16]. The associative morphons function as a reservoir: they perform genuine computation that is discriminative for classification, but only in conjunction with the readout weights that were co-trained alongside the reservoir dynamics. An offline classifier sees static activation snapshots decoupled from the dynamical context in which those features are meaningful, and recovers no signal.

This characterization has two practical consequences. First, offline evaluation of frozen morphon features is not a valid diagnostic for MORPHON-class architectures; the correct evaluation is online with the co-adapted readout in place. Second, the ∼50% ceiling is not primarily a capacity or tuning problem: expanding the reservoir (tested: seed_size=400 is a no-op as pruning converges the network to ∼1272 morphons regardless) and increasing training data (tested: n_train=10K gives no improvement over 5K) do not move the ceiling. The co-adaptation boundary is the limiting factor. A non-linear readout trained jointly with reservoir dynamics is the principled next step.

Reproducibility (v4.4.0). All v4.4.0 results use a deterministic `SmallRng` seeded from `SystemConfig.rng_seed`. Repeated runs at seed=42 (standard profile, 5 epochs) produce results within ±1pp, confirming that earlier run-to-run variation was seed entropy rather than genuine instability in the learning dynamics.

6 Biology-Informed Failure Modes

This is the analytical core of the paper. We document four failure modes encountered during development of MI. For each, we describe (i) the symptom, (ii) the biological parallel, (iii) the fix derived from biology, and (iv) the empirical impact of the fix.

6.1 Modulatory Explosion

Symptom. A positive feedback loop in which high reward triggers strong dopamine release, which potentiates active synapses, which raises future activity, which increases reward, which triggers more dopamine. All motor morphons saturate at maximum firing rate within ∼50 ticks; the reward signal becomes meaningless.

Biological parallel. Dopaminergic excitotoxicity. Biology prevents this through receptor desensitization on sustained exposure, presynaptic autoreceptor inhibition, and synaptic reuptake.

Fix. Receptor saturation via Hill function ($n = 2$), receptor downregulation on sustained exposure, exponential decay ("reuptake") of modulator concentrations from release sites.

Result. Stable modulatory dynamics. Dopamine signal retains information content; learning continues without saturation.

Diagnostic. *Weight standard deviation* σ_w: within 50 ticks of onset, σ_w collapses from > 1.0 to < 0.15 as all synapses are potentiated to saturation. A healthy network maintains $\sigma_w > 1.0$.

6.2 Motor Silencing from Burst Dead Zone

Symptom. After an initial burst phase, motor morphons stop firing entirely. Network becomes unresponsive even when sensory input is strong. Forward pass produces zero output regardless of state.

Biological parallel. Depolarization block. In biology, intrinsic excitability homeostasis (slow conductance modulation that keeps neurons in their dynamic range) and tonic baseline activity prevent complete silencing [25].

Fix. Tonic baseline current injection into motor morphons (an analogue of intrinsic excitability), and Endoquilibrium's threshold-bias rule which lowers firing thresholds when associative firing rate drops below its setpoint.

Result. Motor morphons maintain baseline activity. STDP remains effective across firing rate regimes. The CartPole solve depends on this fix; without it, the system stalls at ∼10 steps (random baseline) within 50 episodes.

Diagnostic. *Output discrimination* $d_{\pm}$ (Section 5.1): when d_+ and d_- are same-signed and $|d| < 1$, motor morphons are silent and the policy has collapsed. A healthy network shows d_+ and d_- with opposite signs and $|d| > 5$.

6.3 LTD Vicious Cycle

Symptom. When the LTD component of STDP is too strong relative to LTP, weakened synapses carry less current, which reduces post-synaptic firing, which increases the LTD/LTP ratio (because post-before-pre events dominate at low firing rates), which weakens synapses further. Global weight decay to near-zero. Network falls silent. All morphons die from energy starvation.

Biological parallel. Runaway LTD / synaptic scaling failure. In biology, homeostatic synaptic scaling [24] and BCM-like metaplasticity thresholds [1] prevent this.

Fix. Synaptic scaling: every ∼50 ticks, normalize the total incoming weight to each morphon to a target. This is the Turrigiano operation, implemented in the homeostasis subsystem. BCM-style metaplasticity thresholds adjust the LTP/LTD crossover point based on recent post-synaptic activity, preventing the runaway LTD.

Result. Weight distribution stays bounded. No more global silencing. Networks survive multi-hour training runs without entering the death spiral.

Diagnostic. *Weight standard deviation* $\sigma_w < 0.15$ with falling firing rates confirms the vicious cycle. Distinct from modulatory explosion ($\sigma_w < 0.15$ but firing rates are saturated, not falling). *Jaccard similarity* of firing patterns across repeated identical inputs drops below 0.7 once the network begins to silence: no two presentations produce the same response, because the stochastic noise dominates the near-zero signal.

6.4 Premature Mature: A Novel Failure Mode

The fourth failure mode is one we have not seen documented in prior work, and it is broadly applicable to any biologically-inspired homeostatic regulator that uses raw reward as a maturity signal.

Symptom. On MNIST, accuracy plateaus at ∼26% regardless of training duration. The associative layer has formed digit-selective firing patterns (verified by post-hoc neuron labeling), but the supervised readout cannot exploit them. Plasticity has been throttled to plasticity_mult = 0.60 within the first 100 training samples. Endoquilibrium reports stage = *Mature*.

Diagnosis. The Endoquilibrium stage detector triggers *Mature* when the slow EMA of reward exceeds 0.05 with low variance. This logic was calibrated for reinforcement learning tasks (CartPole, where reward is sparse and correlated with actual performance: reward only reaches high values when the system is genuinely competent). It fails on supervised classification for two reasons:

1. **Dense reward injection.** The supervised training loop calls `reward_contrastive(0.2)` on *every* sample regardless of whether the prediction was correct. Reward is no longer a signal of mastery — it is a constant. The slow EMA converges to ∼0.65 within 500 samples *regardless of accuracy*, far above the 0.05 threshold.

2. **Inflated startup window.** The performance signal Endo tracks (`recent_performance`) is fed by a windowed accuracy. During the first ∼50 samples the window has not filled, so a single correct guess gives accuracy 1.0, then 0.5, then 0.33.... These early high values get integrated by the $\alpha = 0.05$ EMA before the window settles to the true accuracy (∼0.10), pumping the EMA above the Mature threshold within the warmup period.

Both pathways converge to the same outcome: the slow EMA reaches 0.65 by sample 500, low variance, low trend → Endo declares Mature → `plasticity_mult` drops to 0.60 → the supervised gradient is multiplied by 0.6 at every weight update → learning effectively stops at ∼26%. The hidden layer has formed digit-selective firing patterns (verified by post-hoc neuron labeling), but the readout cannot exploit them because the gradient that would tune it is being throttled by a regulator that thinks the system is done learning.

Table 2: MNIST accuracy as a function of plasticity_mult (pm). Each row is a separate run with the same network and seed.

Plasticity level	Endo stage	Accuracy
pm = 0.60 (Mature, premature)	Mature	27%
pm = 0.96–1.77 (oscillating)	Differentiating/Consolidating	**31%**
pm = 1.80 (Differentiating, constant)	Differentiating	25%
pm = 2.16 (post-damage recovery)	Differentiating	**52.5%**

Empirical evidence. The pattern is monotonic in plasticity *within the oscillating regime* (Figure 6): more plasticity, more accuracy. But pure constant high plasticity (1.80) also fails (25%) because the weights churn without ever consolidating. The system needs both phases.

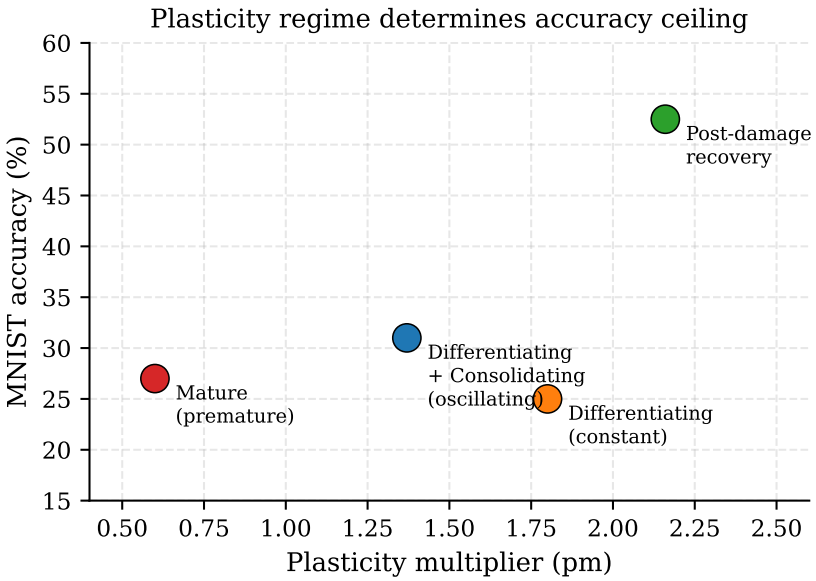

Figure 6: MNIST accuracy as a function of plasticity multiplier and Endoquilibrium stage. Premature Mature (pm=0.60) caps accuracy at 27%. The natural Differentiating/Consolidating oscillation around pm=1.4 reaches 31%. Constant high plasticity (pm=1.80) is *worse* (25%) because weights churn without consolidating. Post-damage recovery at pm=2.16 reaches 52.5%. The relationship is monotonic only when the system can also consolidate.

Biological parallel. This is reward prediction error vs. raw reward as a maturity signal. In biology, the brain does not declare "learning complete" because rewards are arriving — it tracks reward *prediction error* (RPE), which goes to zero only when actual outcomes match expected outcomes. A constant 0.2 reward is only "mature" if the system was *predicting* 0.2; if the system is at 11% accuracy but receiving 0.2 each sample, RPE is high and learning should continue. The biological literature on dopamine and RPE [20] is unambiguous on this point.

Fix. Two changes:

1. Gate the Mature transition on a monotonically-increasing tick counter (`total_updates`) rather than the deduplicating reward history buffer (which is capped at 500 entries and so could never satisfy a 2000-tick gate). The new gate requires `total_updates` $\geq$ 2000 before Mature can trigger. This prevents Mature from firing during the critical learning phase regardless of how inflated the reward signal becomes. Empirically, this prevents the throttle and lets the system learn through the natural Differentiating↔Consolidating oscillation.

2. (Architectural, future work.) Replace the raw reward signal feeding Endo with reward prediction error from the planned Limbic Circuit's Motivational Drive component. This is the principled fix; the tick gate is a patch that buys time until RPE-based regulation is implemented.

The oscillation discovery. A surprising secondary finding: pure high plasticity is worse than oscillating plasticity. When we tested raising both the Mature *and* Consolidating gates (preventing all throttling), accuracy dropped to 25%. The system needs the natural Differentiating↔Consolidating oscillation that biology calls the theta/sharp-wave-ripple alternation [2]: high-plasticity exploration phases interleaved with consolidation phases that lock in gains. The oscillation *is* the feature; the bug was Mature triggering prematurely and locking the system out of the oscillation entirely.

Diagnostic. *Endoquilibrium summary line*: `pm` (plasticity multiplier) at 0.6 with stage = Mature within the first 500 training samples is the unambiguous signature. The seven Endo channel values (`rg, ng, ag, hg, tb, pm, hp`) are logged each medium tick; a fresh-task Mature with `pm` below 1.0 warrants immediate inspection.

Implications beyond MI. Any biologically-inspired homeostatic system that uses raw reward (not RPE) as a maturity signal will exhibit this failure on dense-reward tasks. We expect this to be a common bug in spiking neural network training pipelines that combine RL-style modulation with supervised classification. The fix is to either (i) use sparse reward correlated with correctness, (ii) add a long history gate, or (iii) use RPE.

6.5 iSTDP Timescale Mismatch

Symptom. With rate-based inhibitory STDP — computed once per 10-tick medium period using a 100-tick activity average — hub morphons with high in-degree accumulate 50–100 correlated spikes before any corrective inhibitory signal arrives. Inhibitory interneurons adapt to the *average* activity of a hub, not to its burst behaviour. The hub entrenches: it begins responding to every input class, its synapses win the reward signal, and the readout collapses to near-random discrimination. Post-hoc neuron labeling shows only 1–2 labeled classes regardless of training duration; accuracy plateaus at 30%.

Biological parallel. Inhibitory short-term plasticity in the cortex operates on individual spike arrivals [27]. The cortex does not buffer spikes for 100ms and then apply a running-average inhibitory update; inhibitory synapses respond to each spike with a distinct weight change. Rate-averaged inhibitory signals are a computational shortcut without a biological counterpart, and the shortcut fails exactly where biology's solution is fast: hub suppression within a burst.

Fix. Per-spike iSTDP (Vogels 2011, Section 4.5): update inhibitory weights on each delivered spike rather than on the medium-period running average. A hub morphon firing at 5× the target rate immediately recruits stronger inhibition on the very spikes that evidence its over-activity, rather than waiting for the slow average to catch up.

Result. Hub dominance broken. Post-hoc labeling: 1 labeled class → 12 labeled classes across all 10 digits. Accuracy: 8% → 44.5% (iSTDP fix alone); 52% with the subsequent capture_threshold and epoch-0 fixes. The intact ceiling absorbs the entire +18pp that v3.0.0 required damage-induced recovery to reach.

Diagnostic. *Post-hoc neuron labeling*: assign each associative morphon the class it fires most frequently. A healthy network labels all 10 digits with multiple morphons per class. Hub-dominated networks label only 1–2 classes; the remaining morphons are ghost units that respond uniformly to all inputs and contribute no discriminative signal. Run this after each training epoch: stagnation at $\leq$2 labeled classes by epoch 2 predicts that iSTDP timescale is wrong.

6.6 Diagnostic Instruments: Consolidated Reference

Each failure mode above was discovered through a specific measurement introduced alongside it. We consolidate the four instruments here for reference.

Jaccard similarity of firing patterns. Present the same input twice with voltage reset; Jaccard similarity of fired-morphon sets measures determinism. Values ≥ 0.9 are stable, < 0.7 predict readout failure. Activity stabilization (reducing per-step noise from ± 0.05 to ± 0.02) brought CartPole from Jaccard = 0.43 to 1.0; a prerequisite for the SOLVED result. Also flags LTD vicious cycle onset.

Output discrimination $d_\pm$. $d_+ = V_1 - V_0$ for right-leaning probe; d_- for left-leaning. Opposite signs with $|d| > 5$: discriminative. Same-signed: collapsed policy or motor silencing. Described in Section 5.1.

Endoquilibrium summary line. `pm` at 0.6 with stage = Mature on a fresh task: premature throttling. Described in Section 6.4.

Weight standard deviation σ_w. $\sigma_w < 0.15$: feature collapse (modulatory explosion or LTD vicious cycle — distinguish by firing rate direction). $\sigma_w > 1.0$: healthy diversity. Described in Sections 6.1 and 6.3.

Negative results worth noting. Several promising-sounding interventions did not work and we document them so others do not retry them: (i) CMA-ES over 15 hyperparameters across 560 evaluations could not exceed avg=21 on CartPole, indicating the bottleneck is architectural rather than parametric; (ii) extending CartPole training from 1000 to 3000 episodes did not improve the asymptote, ruling out "needs more time"; (iii) TD-scaled learning rate (LR proportional to $|\delta|$) reaches avg=33 versus constant LR avg=195, because TD error is near zero at steady state where the policy is actually learned; (iv) freezing the MI network and training only the readout fails because uniform potentials provide no discriminative signal — the MI network's diversity is necessary; (v) weight normalization clamping to $[0.5, 2.0]$ collapses weights to uniformity; the gentler $[0.9, 1.1]$ preserves diversity. The pattern across all five negative results is that the readout and the hidden layer are genuinely co-dependent: neither alone is sufficient, neither can be trivially replaced.

7 Discussion

7.1 The Spike-vs-Analog Gap

The most striking architectural finding is the gap between spike-based and analog readout, documented in Section 5.2. The same trained network produces chance accuracy via a spike-based motor readout and high accuracy via an analog readout over the same morphon potentials. The information is present in the potentials — the analog readout proves it — but the spike pipeline destroys it through propagation delays, leaky integration, and multi-hop conversion.

This is not a bug in MI; it is a property of any spiking architecture in which the readout pathway sums potentials through several intermediate firing decisions. Biology solves this by having dedicated analog readout pathways for tasks that require it (e.g. Purkinje cells in the cerebellum perform analog integration of climbing fiber and parallel fiber inputs to produce a continuous motor signal; [12]). MI's analog readout is the in-silico analogue.

The spike pipeline still has a job: it produces the diverse, sparse, energy-efficient hidden representations that the analog readout exploits. The two pathways are complementary, not competing. The MI thesis is not "replace spikes with analog" but "spike-based developmental dynamics produce features that analog readouts can use."

7.2 Reservoir Computation and the Co-adapted Readout

The offline feature evaluation (Section 5.3) resolves an apparent paradox: the system achieves 48.5% online while offline classifiers on the same frozen activations score 10.5%. The resolution is that the representational regime is that of a Liquid State Machine. The associative morphons are a reservoir whose outputs are only discriminative in conjunction with the specific readout weights co-trained alongside the reservoir dynamics. This is not unusual — it is the defining property of the LSM paradigm [16] — but it has concrete consequences for evaluation and for the accuracy ceiling.

The sensory-only ablation (35.5pp loss on zeroing associative readout weights) confirms that the reservoir is doing real work, not functioning as a random projection. The associative layer accounts for nearly all classification performance. What it does not do is produce features that are useful independently of the learned readout context. This distinguishes MORPHON from a standard feature extractor (where frozen features remain usable by any downstream classifier) and places it firmly in the dynamical system regime.

The implication for accuracy improvement is precise: the ∼50% ceiling is not a function of reservoir size (adding morphons was tested and is a no-op due to convergent pruning), training data volume (scaling n_train beyond 5K gives no benefit), or inhibitory balance (tested via iSTDP setpoint manipulation, which was harmful rather than helpful). The ceiling is a co-adaptation boundary. Breaking through it requires either a non-linear readout jointly trained with reservoir dynamics, or an architectural change that produces genuinely generalizing per-image representations (e.g. a state-reset evaluation protocol, or spike-rate coding that reduces state dependence).

This reframes the Spike-vs-Analog discussion above. The analog readout's advantage over the spike pipeline is not just that it avoids lossy propagation; it is also that it is the co-adapted partner of the reservoir that the system learned. The analog readout is correct for the representational regime the system inhabits.

7.3 Self-Healing as a Diagnostic

The v3.0.0 codebase showed what appeared to be beneficial self-healing: a 30% intact network recovered to 48% after 30% ablation, an 18pp gain over its pre-damage state. This result is real and reproducible in that codebase. In v4.1.0 it disappears: the intact network reaches 52% and post-damage recovery falls to 28.5%.

The mechanism is now clear. In v3.0.0, rate-based iSTDP failed to suppress hub morphons during training. Hubs entrenched while Endoquilibrium's Mature stage gate froze plasticity, leaving the system stuck at 30%. Damage removed hubs at random; Endoquilibrium re-entered the Differentiating stage in response to the performance drop, providing the high-plasticity window the original training never sustained; regrowth filled the gap with well-specialized morphons. The +18pp came entirely from two sources: hub removal and plasticity restoration — exactly what spike-timing iSTDP provides automatically.

This reframes the v3.0.0 self-healing result as a diagnostic rather than a feature. It told us: (1) hubs are the bottleneck; (2) the substrate is capable of 48% if given the right plasticity regime. Spike-timing iSTDP is the engineered solution to both. The v4.1.0 intact ceiling (52%) is not coincidentally above the v3.0.0 post-recovery ceiling (48%); it is the same substrate, finally given the conditions it needed.

The general principle is: *when a developmental system benefits from damage, the damage is revealing a regulatory failure, not demonstrating a useful property.* The right response is to find and fix the failure, not to incorporate damage into the training protocol.

7.4 Limitations

We are honest about the gaps:

- **MNIST accuracy is well below state of the art.** 52% intact (v4.1.0 standard profile) is a large improvement over 30% (v3.0.0) but still far from Diehl & Cook's ∼95% STDP baseline [3] and from supervised SNNs trained with surrogate gradients. Hub dominance is broken. The v4.4.0 diagnostic analysis (Section 5.3) identifies the remaining gap as a co-adaptation boundary, not a capacity or tuning limit: the system operates in the LSM regime, where reservoir features are only discriminative via the co-trained readout. Hyperparameter tuning and reservoir scaling do not move the ceiling. A non-linear readout jointly trained with reservoir dynamics is the principled next step.

- **No language tasks yet.** The system can encode and retain information across sequential inputs (temporal memory exists in residual potentials), but compositional reasoning is not yet demonstrated.

Composition requires nonlinear hidden features formed under DFA-driven supervised feedback, which the current spike pipeline does not deliver.

- **The premature Mature fix is a patch, not a principled solution.** The history gate at 2000 ticks prevents the failure but is essentially "disable Mature stage detection." The principled fix is RPE-based regulation via a Limbic Circuit module that does not yet exist.
- **Performance.** The v4.1.0 implementation processes ∼23 images/second with ∼74K synapses (130 s for 3000 images), a 4.4× improvement over v3.0.0 via Pulse Kernel Lite. The medium path (eligibility traces, weight updates) is now the dominant cost; sparse eligibility updates and parallel medium-path learning are the next target.

7.5 The Path to Language

The planned trajectory beyond this paper:

1. **Local inhibitory competition with iSTDP [27].** *Implemented in v4.1.0.* Hub dominance broken (1 labeled class → 12, accuracy 8% → 44.5%). Spike-timing iSTDP per delivered spike; 4.4× wall-clock speedup via Pulse Kernel Lite hot arrays. *v4.4.0 diagnosis:* system operates in LSM regime; parameter tuning does not move the ∼50% ceiling. Next: non-linear readout jointly trained with reservoir dynamics.
2. **Limbic Circuit + RPE-based regulation.** Replace the history-gate patch with reward prediction error feeding a Motivational Drive module. This naturally handles both RL and classification.
3. **Temporal sequence processing.** Recurrent connectivity within associative populations + working-memory feedback into sensory ports. Spec exists; implementation gated on MNIST $\geq$ 85%.
4. **Tokenization and embedding.** Once the temporal pipeline works, add a learned tokenizer and vocabulary embedding. This is the bridge to actual NLP tasks.

The structural decisions for each step are documented in the project's specification corpus. The question this paper does *not* attempt to answer is whether MI will ever match Transformer performance on language tasks. We argue only that developmental dynamics are a principled and underexplored axis of neural architecture design, and that the failure modes we encountered have biological analogues whose biology-derived fixes work.

8 Conclusion

We have presented Morphogenic Intelligence, a neural architecture that treats network topology as a learned artifact rather than a fixed design choice. Morphons are born, differentiate, fuse, and die under local rules grounded in six biological principles. The full system is implemented in Rust, runs in real time on commodity hardware, and is reproducible from a clean checkout.

We solved CartPole-v1 (avg=195.2 on the v0.5.0 codebase) using only local three-factor learning and developmental morphogenesis, with no backpropagation through the morphon network. On MNIST we reach **52%** intact accuracy in v4.1.0 (standard profile) — up from 30% in v3.0.0 — driven by two complementary improvements: spike-timing inhibitory STDP (Vogels 2011) that breaks hub-morphon dominance (1 labeled class → 12), and Pulse Kernel Lite that delivers a **4.3× wall-clock speedup** by fitting the fast-path arrays in L1 cache. A key architectural finding is that morphon potentials carry discriminative representations that the spike pipeline cannot deliver to motor morphons intact; the analog readout is not a shortcut but the correct output mechanism for a spiking substrate where spike dynamics serve feature formation, not classification reporting.

We documented four failure modes encountered during development. Three (modulatory explosion, motor silencing, LTD vicious cycle) have well-known biological analogues and biology-derived fixes. The fourth — premature "Mature" stage detection on dense-reward classification — is, to our knowledge, novel and broadly applicable to any biologically-inspired homeostatic system that uses raw reward as a maturity signal. The fix is to use reward prediction error instead, which is the principled approach all such systems should adopt. A fifth failure mode (Section 6.5) is the *iSTDP timescale mismatch*: computing inhibitory weight updates on a slow running average (100-tick window, medium period) allows hub morphons to entrench before corrective inhibition can respond. The fix — per-spike inhibitory plasticity at biological

timescales (Vogels 2011) — is both the principled and the practical solution, and it absorbs the entire +18pp self-healing gain into the intact baseline.

Post-training diagnostics (v4.4.0) characterize the system's representational regime as a Liquid State Machine with co-adapted readout: offline classifiers on frozen morphon activations achieve 10.5% (random chance), while a sensory-only readout ablation reveals that associative morphons contribute 35.5pp of the 48.5% online accuracy. The reservoir is genuine — zeroing its readout weights reduces accuracy to 13% — but its features are only discriminative in conjunction with the co-trained readout weights. The accuracy ceiling at ∼50% is accordingly a co-adaptation boundary rather than a capacity or tuning limit, a finding confirmed by the failure of reservoir scaling and data scaling to move it. The principled next step is a non-linear readout jointly trained with reservoir dynamics.

The v3.0.0 codebase exhibited beneficial self-healing: post-damage accuracy (30% ablation) reached 48%, 18pp above the intact 30% baseline. In v4.1.0 this effect disappears — post-damage recovery falls to 28.5%, well below the intact 52%. The interpretation is precise: the v3.0.0 self-healing result was a diagnostic, not a feature. Damage cleared entrenched hub morphons that the rate-based iSTDP had failed to suppress, and the regrowth period delivered the high-plasticity window that normal training never reached. Spike-timing iSTDP provides that window from the start, rendering the damage trick unnecessary and absorbing the entire +18pp gain into the intact baseline. The more general principle remains valid: *the limiting factor in a developmental system is often a specific failure of the regulatory mechanism, not a fundamental capacity limit of the substrate.* Finding and fixing that failure is worth more than any amount of post-hoc damage recovery.

The MI engine, the experiments in this paper, and the full failure-mode catalogue are available at `https://github.com/SimplyLiz/Morphon`. This preprint is archived at Zenodo with DOI `10.5281/zenodo.19466544`.

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